

Optimized Machine Learning Framework for Dental Caries Detection

Architectural Document

1 Architecture Overview

The proposed system is designed to detect dental caries from panoramic dental radiographs (OPG X-rays) using deep learning. The system follows a **Monolithic Architecture**, where all components including preprocessing, training, evaluation, and prediction are integrated into a single pipeline.

This architecture is suitable for standalone machine learning applications without distributed services.

2 Application Architecture

The system consists of the following modules:

- **Input Module:** Accepts dental X-ray images
- **Preprocessing Module:** Resizes images to 224x224 and normalizes them
- **Model Module:** ResNet-50 based deep learning classifier
- **Prediction Module:** Outputs Caries or NonCaries
- **Evaluation Module:** Calculates performance metrics

3 Architecture Type Selection

- Microservices Architecture: Not used
- Event-Driven Architecture: Not used

- Serverless Architecture: Not used
- **Monolithic Architecture: Selected**

4 Database Design

4.1 ER Diagram

The ER diagram illustrates the relationships between images, labels, model information, and predictions in the system.

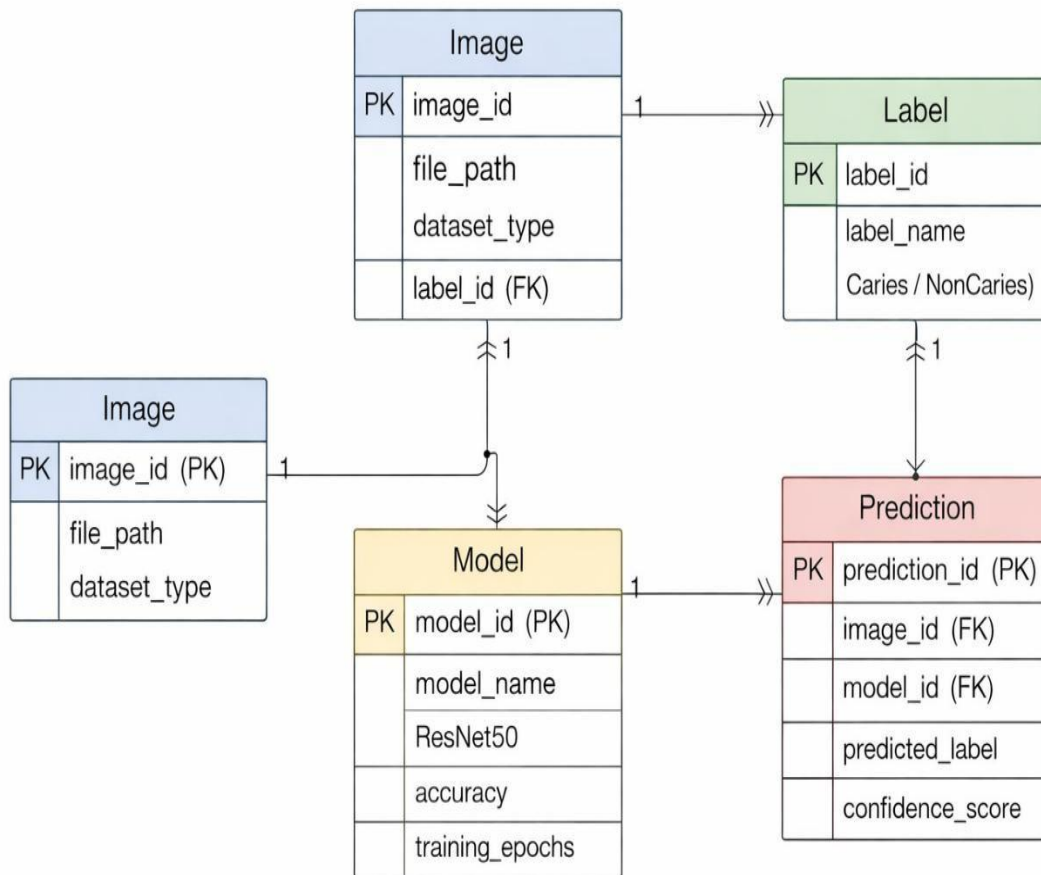


Figure 1: Entity Relationship Diagram

4.2 Schema Diagram

The schema diagram represents the database tables, their attributes, and relationships including primary and foreign keys.

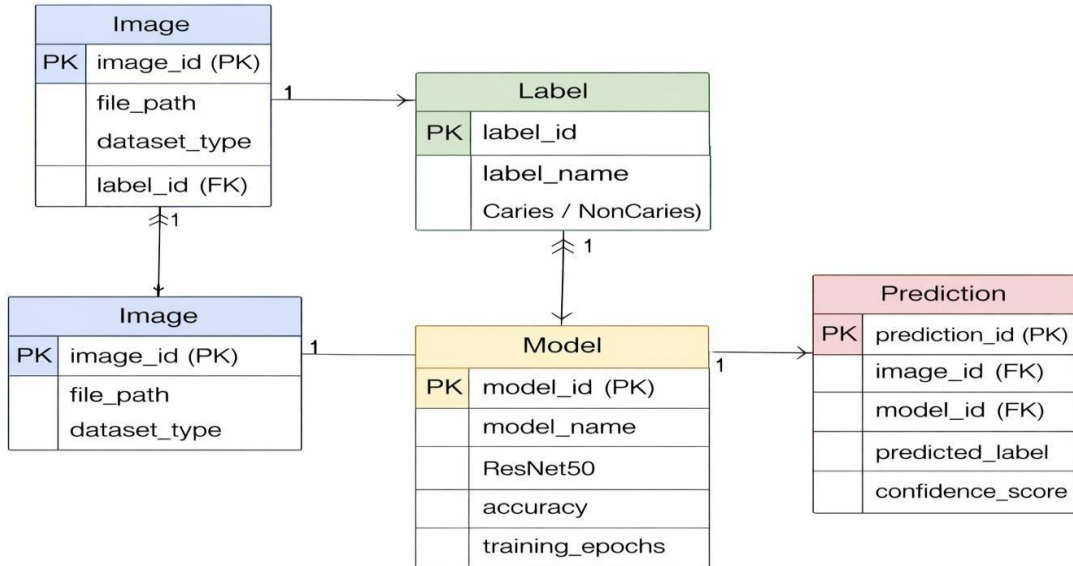


Figure 2: Schema Diagram

5 Data Exchange Contract

The Data Exchange Contract defines how data flows between different components of the system.

5.1 Data Sets

- Dental OPG X-Ray Dataset
- Total Images: approximately 9481
- Classes: Caries and NonCaries

5.2 Frequency of Data Exchange

- Data exchange occurs during training and evaluation phases
- No real-time data exchange is involved

5.3 Mode of Exchange

- File-based data exchange (image dataset)
- No API or queue-based communication

6 System Workflow

1. Input dental X-ray image
2. Preprocess image (resize and normalize)
3. Feed image into ResNet-50 model
4. Generate prediction (Caries / NonCaries)
5. Store prediction results

7 Hybrid Pipeline (Experimental)

An experimental hybrid pipeline was implemented:

- YOLO used for region detection
- Detected regions are cropped
- Cropped regions are passed to ResNet-50 for classification

However, due to lack of domain-specific training data, the detection model did not perform effectively, leading to reduced accuracy.

8 Conclusion

The system successfully implements a monolithic deep learning architecture for dental caries detection. The ResNet-50 model achieves strong performance and serves as the core component of the system. The hybrid pipeline provides insight into potential future improvements.